Rendered, Not Transported: Epistemic Disposition in Multi-Agent LLM Pipelines

Sam Bobo

Draft v0.7 — August 2026 | Target: arXiv cs.CL

Abstract

In a multi-agent LLM pipeline, several models contribute and a final model writes the answer the user reads. Qualifications the specialists wrote— labelled assumptions, hedges, notes that information may be out of date— often do not survive into that answer. The usual explanation is that they are lost when one component hands off to the next, which has motivated proposals to carry uncertainty in a separate channel alongside the text. That explanation does not fit the common case. Holding the pipeline, the models, and the specialists' text fixed, and changing only the instructions given to the final model, moves the amount of qualification in the output by three to four times. Nothing before that step differs between the conditions, so nothing was lost on the way: the qualifications were in the text, and the final model simply did not write them out.

Across roughly 1,300 pre-registered runs of a four-model council on one consumer machine, we find that multi-agent structure by itself contributes almost nothing—three specialists merged without the right instruction match a single model working alone, and adding rounds of debate does not help—while a well-prompted single model matches the whole council for sheer quantity of qualification. What the council alone achieves is knowing *when* to qualify, because its instruction depends on what the specialists actually flagged. Training this behavior into model weights failed everywhere we tried it: on the specialists, on the final writer, with three optimizers, at triple the data, and with a reward computed on the pipeline's own output. Instructions work instead, and work precisely—each clause lifts the behavior it names and no other. Given a neutral prompt, four different architectures produce nearly identical amounts of qualification, so what separates candidate models is not disposition but how strongly each responds to being asked.

1 Introduction

Production LLM systems increasingly take the form of pipelines: multiple models, often specialized, whose outputs are aggregated by a final model into the answer a user sees. A recognised failure of such systems is that epistemic qualification does not survive the trip. A specialist writes "assuming 60% adoption" or "this may have changed since my training data"; the user reads a confident synthesis with the qualification gone.

Where does the qualification go? The prevailing answer places the loss at the boundary between agents. On this account, the interfaces that connect components pass along decisions but not the uncertainty behind them, so a caveat that existed inside one model is gone by the time the next one reads its output. The proposed fixes follow: pass numeric confidence estimates alongside each step [27], or attach a separate uncertainty-bearing representation to every handoff, on the grounds that ordinary text cannot preserve it [26]. Both remedies require changing the system's architecture and reading model internals, which rules them out for components accessed as black boxes.

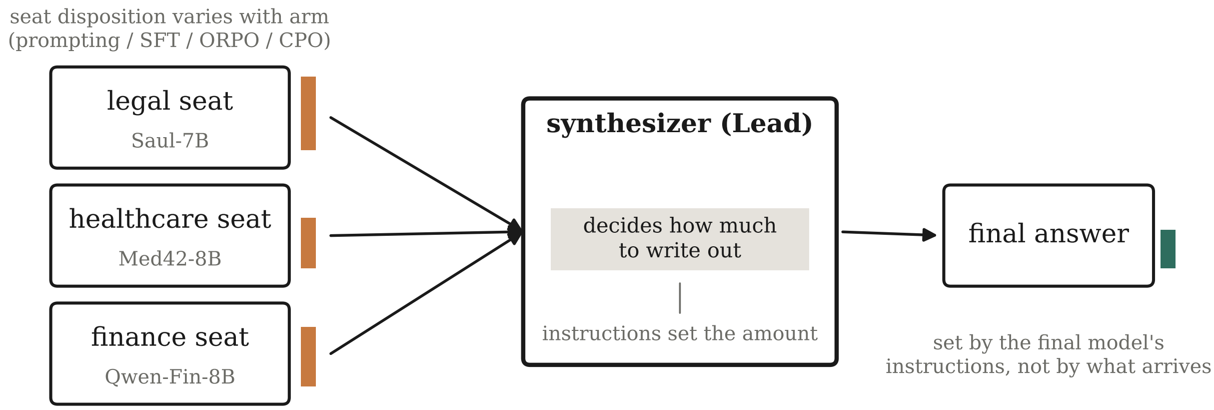

Figure 1: How much qualification the specialists produce varies a great deal with how they were trained (orange bars). How much reaches the finished answer (green) barely varies at all: the final model decides that, and its instructions decide how much it decides on (§6, Fig. 4).

Our measurements point somewhere else. Keep the pipeline, the models, and the specialists' text fixed, and change only the instructions the final model receives. The amount of qualification in the output moves from 0.16 to 0.53 as preservation clauses are restored to that model's prompt (§6), and from 0.20 to 0.64 when a single clause is added (§9). Nothing before the final step differs between those conditions. If an instruction to the last model can bring a qualification back, the qualification was still there to be found—so it was not destroyed earlier. What varies is whether the final model writes it out.

We take this to mean that qualification is decided at the point of writing rather than in the plumbing, and that the practical remedy is much cheaper than a new channel: an instruction, given in one place, to a model whose internals need not be visible. The narrower point deserves stating. Some interfaces really do discard uncertainty—a structured tool call with fixed fields has nowhere to put it, and the work cited above is right about those. Our claim concerns the case where one model's prose is read by another, which is how specialist output usually reaches an aggregator. There, the information arrives intact and is dropped when the answer is written.

Two axes. We distinguish **content**—whether the right substantive material appears, measured by per-case expert rubrics—from **disposition**, what a model chooses to emit about its own claims. Our experiments show these are orthogonal: the model family that wins content loses disposition and vice versa. This paper is about the second axis, and about the fact that in a pipeline it is governed almost entirely by the final stage.

Contributions. We report a test that separates the two explanations above, holding the specialists' text fixed and varying only what the final model is told (§6, §9). We compare six pipeline designs with the model held fixed, separating what multi-agent structure contributes from what a conditional instruction contributes (§4). We compare three ways of training the behavior in, at both ends of the pipeline, under predictions registered in advance (§5), and characterise the final model's behavior as a function of what it receives and what it is told (§6, §9). We show that models differ far less in what they produce unprompted than in how strongly they answer an instruction (§8). Throughout, conclusions are checked against three independent measurement instruments, one of which caught a result the primary measure had produced on its own (App. E). Everything ran locally at no API cost, with full logs, pre-registrations, protocol amendments, refuted predictions, and an audit that withdrew one arm after publication of its figure.

2 Related work

Multi-agent orchestration. Debate and role-played collaboration improve factuality and reasoning [1, 2, 3, 4], and recent work compares orchestration against the strongest single model [5] or trains orchestrators end-to-end [6]. A 2026 line now reports that well-prompted single agents match or beat multi-agent systems under matched token budgets [33]; our architecture comparison reproduces that negative result on a new axis (disposition *magnitude*) and then identifies an axis where orchestration does pay—*calibration*—which that literature does not measure. This literature measures content outcomes. The closest work on the *aggregation* step, Parallel-Synthesis [7], trains a synthesizer adapter and explicitly distills reasoning behavior from text-concatenation synthesis—an implicit acknowledgment that naive aggregation loses behavior—which our rendering account explains and quantifies for epistemic behavior specifically, and locates in the writer's instructions rather than in the concatenation.

Specialists vs. generalists. Evidence is mixed on whether small domain fine-tunes beat larger generalists in-domain [8, 9]. Our content results replicate the negative side (a 20B open MoE outperforms a four-specialist council $42\% \rightarrow 25$–31% on rubric coverage) and motivate the pivot: the surviving specialist value is dispositional, not informational.

Uncertainty expression and abstention. Verbalized uncertainty [10], linguistic calibration [11], knowing-what-you-know [12], abstention training [13] and recent confidence-aware abstention work [14, 15] train and evaluate single models end-to-end. A parallel line asks whether epistemic markers *faithfully* track a model's internal confidence—and finds systematic deficiencies [16, 17, 18]. Our contribution is orthogonal to faithfulness: we take marker *emission* as the observable and ask where it originates and survives in a pipeline, not whether it is calibrated to hidden confidence. A seat's trained uncertainty behavior must survive an aggregator, and mostly does not. The closest single-model neighbor is certainty distortion [24]: language models rephrasing scientific and clinical text systematically fail to preserve the source's expressed certainty, distorting up to 75% of outputs with a bias toward *overstating* confidence, compounding under repeated paraphrase, and only partly fixed by prompting. A clinical benchmark reports the same failure for diagnostic uncertainty under summarization [25]. Both study one model transforming text, and neither attempts to train the behavior in. We differ on three counts: our setting is a pipeline in which distinct specialist models feed an aggregator; we test installation by prompting, SFT, ORPO and CPO at both ends, and find weights move nothing; and we characterise the aggregator itself, which does not merely attenuate what it receives but can produce less when given more. Their finding that prompting partly helps and our finding that instructions are the whole lever converge from different directions, which we take as mutual corroboration.

The transport account, and where we disagree. Two 2026 lines make the pipeline dimension timely and both adopt a transport framing. "Confidence laundering" [26] argues, as a position, that standard agent interfaces perform a many-to-one projection from uncertain internal states to discrete commitments, so that uncertainty is *structurally erased by the commitment format itself*; its remedy is a latent uncertainty carrier attached to each handoff, on the reasoning that text cannot preserve pre-commitment fragility. A quantitative line propagates *numeric* confidence estimates through multi-step agent decisions [27].

We agree with the phenomenon and dispute the locus. Their argument is strongest where the interface is a tool call or a schema-constrained field, which genuinely cannot represent a distribution. It is weakest exactly where aggregation usually happens: prose. Our ablations hold the prose fixed and vary only the reader's instructions, and recover 3–4$\times$ swings in emitted disposition (§6, §9). If the qualifications had been destroyed before the final step, no instruction to that step could bring them back; they do come back, so they were still there. The implication is practical as well as conceptual: where the transport account prescribes an architectural change requiring access to model internals, the rendering account prescribes an instruction, which works on black-box components and costs nothing. We regard the two as complementary rather than exclusive—different interfaces, different dominant failure—but the distinction is testable, and we report the test.

Rendering beyond epistemics. Independent evidence for a register-like effect comes from stylistic rewriting: [32] finds that LLMs revising personal narratives shift all thirteen measured linguistic markers in the *same direction* across three frontier models—toward a more polished, less situated register—regardless of user intent, and that voice-preserving prompts reduce the effect magnitude by 32% without changing its direction. That is our mechanism in a different behavioral domain: a rewriter renders in its own band, and instructions are a partial counter-lever. Their per-model effect sizes also differ ($|d| = 0.60$, 1.29, 1.45), consistent with our finding that models differ in instruction *gain* rather than in baseline emission. We differ in object (epistemic disposition, not voice), in setting (a multi-agent pipeline, not single-stage rewriting), and in testing whether the band can be moved by training at all—which they do not attempt.

Preference optimization for behavior. DPO [19] and ORPO [20] are standard for style/behavior shaping; DPO's length bias is documented [21], which our pair-construction controls address. Preference-data scaling is itself contested: simply adding more samples per prompt often fails to improve and can degrade downstream behavior [22, 23]—consistent with our dose-invariance null, which we further attribute to a downstream aggregation ceiling rather than to the optimizer. Comparisons of system-prompt steering vs. preference training exist for single models; we compare them *through* a pipeline.

3 Apparatus

Council. Four local models via Ollama on a 32 GB MacBook Air M5: Phi-4-14B Lead (planner + synthesizer), and three specialist seats—Llama3-Med42-8B (healthcare), Saul-7B-Instruct-v1 (legal), Qwen-Open-Finance-R-8B (finance), all Q4/Q8 GGUF. The Lead performs step-back decomposition [34] and dispatches self-contained sub-questions; seats never see the original query (this architectural choice, not prompt strengthening, eliminated cross-domain lane bleed at sub-10B scale). The Lead then synthesizes under a structurally enforced Tensions-then-Synthesis prompt containing four PRESERVE instructions.

Cases. Seven three-domain scenarios: five failure-mode-targeted (synthesis stress, jurisdictional vocabulary, quantitative discipline, recency honesty, adversarial cross-domain tension) plus a matched disposition pair—case 6 (*trigger-heavy*: demands all five behaviors simultaneously) and case 7 (*trigger-light*: an organizational-strategy question warranting none). Case 7 is the **responsiveness gate**: any arm emitting disposition there is behaving habitually, not responsively.

Scale. 462 imported, audited runs (append-only JSON, one file per run, each carrying per-phase inputs/outputs/backends); all inference via Ollama on one fanless M5 MacBook Air, 32 GB unified memory, 26.8 GB Metal working set. Runs are nondeterministic across repeats even at temperature 0 (runtime batching); we treat repeats as seeds and report bootstrap intervals.

Metrics. Behavior density = pattern-matched occurrences of the five behavior families (four detectable; see below) per 1,000 characters. CDS $=$ density $\times \sqrt{\text{distinct behaviors}/5}$. ALR $=$ council density/matched single-shot density. Seat-level density is computed on the specialist's own turn from the audit log; final density on the synthesized output. (Definitions and regexes are versioned with the code; see §10.) The behaviors are not spread evenly across specialists: disclosing a training cutoff is near-universal, flagging modelled assumptions lives almost entirely with the finance specialist, and distinguishing terms of art is too rare to measure anywhere. Per-behavior claims are therefore tested on the specialist that actually produces the behavior (App. F). Which behavior dominates the finished answer, by contrast, turns out to be a property of the final model rather than of the specialists (§9).

Table 1: Architecture comparison (Fig. 2). Same model in every role; bootstrap 95% CIs over $n = 30$ trigger-case and $n = 5$ trigger-light runs per arm.

Architecture	signal / instruction	trigger density	trigger-light gate
single-shot	none / none	0.14 $[0.09, 0.19]$	0.00 $[0.00, 0.00]$
flat merge	3 agents / naive	0.16 $[0.11, 0.22]$	0.06 $[0.02, 0.11]$
debate (2 rounds)	3 agents+revision / naive	0.16 $[0.12, 0.20]$	0.05 $[0.00, 0.10]$
self-refine	self-critique / none	0.08 $[0.05, 0.12]$	0.00 $[0.00, 0.00]$
single + spec	none / unconditional	0.50 $[0.41, 0.60]$	0.15 $[0.08, 0.22]$
council	3 agents / **conditional**	$\mathbf{0.57}$ $[0.45, 0.71]$	$\mathbf{0.00}$ $[0.00, 0.00]$

Installation arms (legal seat, single-variable design, seed 42): A$'$ = conversion-control baseline (identical Saul weights through our fp16→GGUF→Q4 pipeline); B = a frozen behavior-spec addendum on the seat's system prompt (five behaviors, one example each, an explicit "do not force it" clause); SFT = LoRA on 91 behavior-rich *chosen* responses; ORPO = LoRA reference-free preference optimization on the same 91 chosen responses paired with behavior-stripped rejections. Pairs are content-controlled (chosen/rejected are rewrites of the same base answer), length-ratio filtered (0.8–1.4) against length bias, overlap-filtered (Jaccard ≥ 0.35 on capitalized tokens), and leakage-screened against all seven cases. Replication arms repeat the recipe with one variable changed: ORPO on dose-matched 91-pair sets for the healthcare seat (Med42, preference-aligned) and finance seat (Qwen-Open-Finance, SFT-only), a CPO arm on the identical legal pairs (only the objective differs), and an ORPO arm on the *synthesizer itself* (Qwen2.5-7B Lead, 88 synthesis-level pairs built from recorded deliberations; sequence length 4,096 — an amendment inherent to the locus, since syntheses integrate all seat outputs and no synthesis example fits the seat arms' 1,792). All protocol amendments were documented before training; predictions were pre-registered before each cell ran.

4 Result 1: what orchestration actually buys

Multi-agent pipelines are widely assumed to improve epistemic quality, and our own first pass supported it: council configurations emit 3–9× the behavior density of matched single-shot baselines (aggregate CDS: specialists-v2 0.928, Opus-as-council 0.669, specialists-v1 0.584, gpt-oss-as-council 0.460, versus 0.159 and 0.099 for the two single-shot arms). That comparison, however, confounds two things: the multi-agent *topology* and the synthesis *prompt* that accompanies it. Cell 6c separates them from one direction—stripping the preservation clauses collapses the full council to 0.16, *below* single-shot—so we ran a direct architecture comparison.

Three pre-registered predictions, all borne out. **Magnitude is not the council's edge** (P8.1): a single prompted model reaches 0.50 against the council's 0.57, CIs overlapping. **Multi-agent structure alone buys nothing** (P8.3): flat merge—the same three specialists with a naive "merge these" prompt—sits at 0.16, statistically indistinguishable from single-shot and disjoint from the council. **The council's edge is calibration** (P8.2): its trigger-light gate is 0.00 $[0.00, 0.00]$ (zero unwarranted disposition on every one of five runs) against the prompted model's 0.15 $[0.08, 0.22]$, disjoint intervals.

The exploratory arms sharpen the picture. Two-round debate, where each specialist revises after reading the others, lands at $0.16/0.05$—identical to a one-shot naive merge: *more agent interaction contributes nothing*. Self-refine is the weakest arm on the board (0.08, below plain single-shot): draft-critique-revise compresses

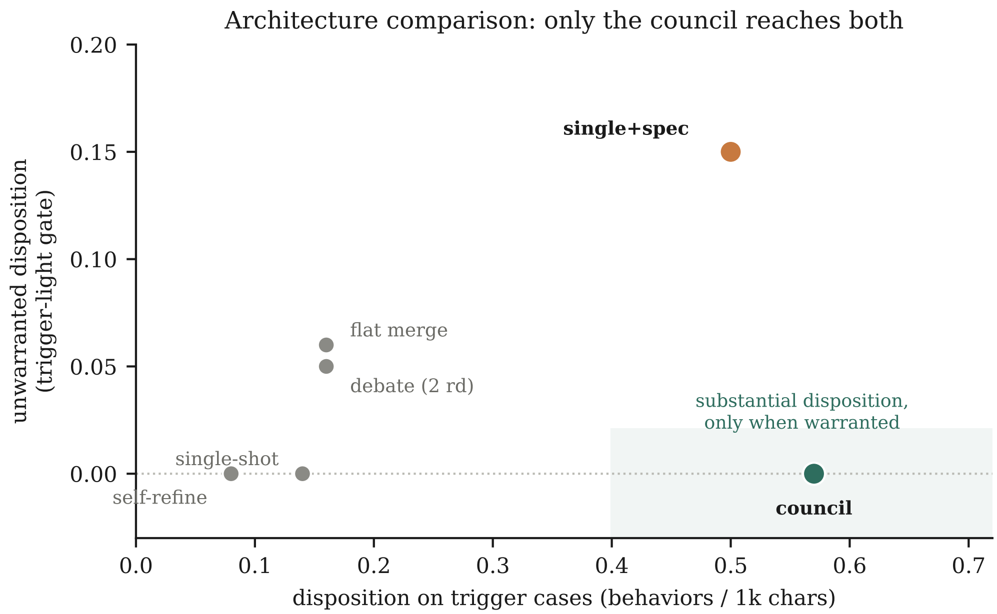

Figure 2: Six orchestration architectures, gpt-oss-20B in every role so shape is the only variable (210 runs, 7 cases $\times$ 5 seeds). The x-axis is disposition where it is warranted; the y-axis is disposition where it is *not*. Multi-agent structure alone (flat merge, debate) barely leaves the single-shot corner; an unconditional instruction (single+spec) buys magnitude but fires when nothing warrants it; self-refine actively removes disposition. Only the council occupies the shaded quadrant.

prose and firms up claims, removing disposition rather than adding it.

The pattern is a conjunction, not a property of any component. Disposition that is both substantial and appropriately gated requires (a) real specialist signal and (b) an instruction that depends on it—*if* a specialist flagged uncertainty, carry it through. Specialists without such an instruction (the flat merge, the debate) produce no more than one model working alone. The instruction without specialists (the prompted single model) produces the quantity but applies it everywhere, since there is nothing for it to check against. Only the council has both, which is why only the council is both substantial and selective.

The gap widens on the hardest case, which demands several kinds of qualification at once: the council arrangements rise to 1.66–$1.82\times$ their usual level there, while a single model falls to $0.68\times$. Asked to do several things at the same time, one model spreads itself thin; the council, with each specialist responsible for less, does not.

5 Result 2: can the behavior be trained in?

If instructions to the final model matter this much, the obvious next question is whether the behavior can instead be built into a model's weights, so that it no longer depends on remembering to ask. We tried, at both ends of the pipeline and by three different methods. The short answer is that it can be put into a specialist and cannot be got out again.

Three measurements are needed to tell the difference between a method that works and one that only appears to. The first is whether the specialist itself changes. The second is whether that change reaches the finished answer. The third is whether the behavior is discriminating—we include one question that warrants no

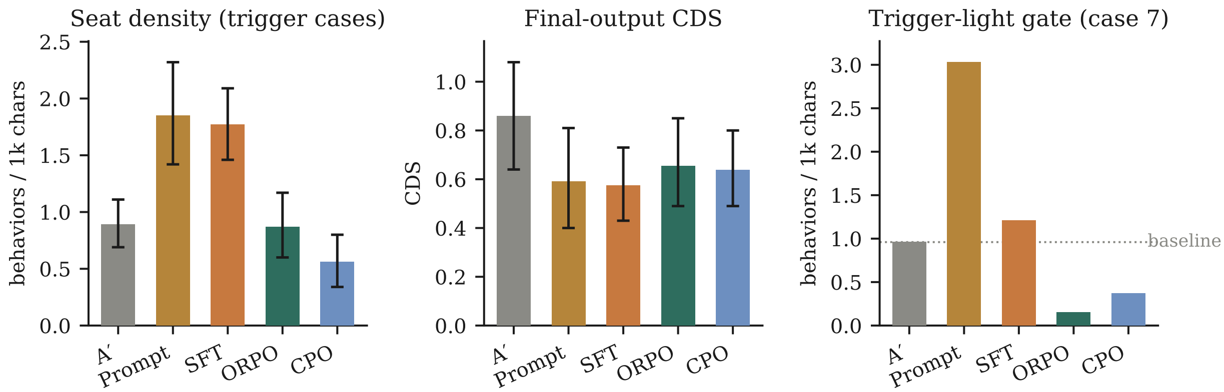

Figure 3: Three measurements on the same arms. Prompting and supervised fine-tuning genuinely change the specialist (left), but the change does not reach the finished answer (centre) and it fires on the question that warrants no hedging (right). Preference optimization changes nothing on the first measure and reduces the unwarranted hedging on the third. Bars: means over $n = 30$ run-level observations (6 trigger cases $\times$ 5 seeds); whiskers: bootstrap 95% CIs (10^4 resamples); gate panel: $n = 4$–5 per arm, dotted line = untrained baseline.

Table 2: Installation arms on the legal seat (cell-2 variance pass; $n = 5$ seeds $\times$ 7 cases per arm, bootstrap 95% CIs over run-level observations). Bold marks CIs clear of baseline (seat) and the best value per column.

Arm	Seat density	Final-output CDS	Trigger-light gate
A′ baseline	0.89 $[0.69, 1.11]$	$\mathbf{0.86}$ $[0.64, 1.08]$	0.96
Prompting	$\mathbf{1.85}$ $[1.42, 2.32]$	0.59 $[0.40, 0.81]$	3.03 ✗
SFT-on-chosen	$\mathbf{1.77}$ $[1.46, 2.09]$	0.58 $[0.43, 0.73]$	1.21 ✗
ORPO	0.87 $[0.60, 1.17]$	0.66 $[0.49, 0.85]$	$\mathbf{0.15}$ ✓

hedging at all, so a model that qualifies everything is visibly distinguishable from one that qualifies the right things. Figure 3 and Table 2 report all three across 140 runs.

Prompting and SFT install real seat-level magnitude (CIs exclude baseline)—and both fail the gate, hedging on the trigger-free case (prompting despite its explicit "do not force it" clause). Both also lose their gains at the pipeline output: final CDS lands at or below the untrained baseline. ORPO installs no significant seat-level magnitude; its effect is qualitative—it is the only arm that *suppresses* unwarranted hedging below baseline while leaving trigger-case behavior intact and imposing no content tax (rubric coverage identical to A′).

The null is dose-invariant. Retraining ORPO at $3.2\times$ dose (292 content-controlled pairs, epoch-matched to the 91-pair run) moved seat-level magnitude by zero: seat density 0.84 $[0.58, 1.11]$ vs. the 91-pair 0.85 and baseline 0.84, even though the larger set raised training-time preference accuracy from 0.48 to 0.94. More preference data was internalized but not emitted—strong evidence that ORPO's seat effect is not gradual installation of magnitude but a bounded suppression of unwarranted behavior. (The suppression itself weakened slightly with dose: the trigger-light gate rose from 0.15 to 0.49, still below baseline; more diverse pairs modestly broadened where the model deploys hedging.)

The null generalizes across lineages; the suppression does not. Pre-registered replications trained ORPO identically on two further seats. On the healthcare seat (Med42, already preference-aligned): no magnitude

(1.40 $[1.15, 1.68]$ vs. A$'$ 1.30 $[1.07, 1.54]$) and no responsiveness effect to detect—the untrained baseline already emits 0.00 on the trigger-light case (dispatched in all runs); a pre-aligned model leaves no miscalibration slack to suppress. On the finance seat (Qwen-Open-Finance, SFT-only, Qwen-3): no magnitude (1.32 vs. 1.37, CIs overlapping) and *no suppression* (gate $0.81 \rightarrow 1.21$ under the pattern metric)—the replication prediction was falsified. The suppression is therefore conditional on base-model miscalibration and not guaranteed even then.

The suppression is method-general; the selectivity is ORPO's. A CPO arm [28] on the identical legal pairs (only the objective changed) reproduces the gate suppression under both instruments (0.37 pattern / 0.45 NLI-cutoff vs. A$'$'s 0.96 / 1.06)—but CPO also dampens *warranted* trigger-case hedging ($0.89 \rightarrow 0.56$), where ORPO leaves it intact (0.87). Gate-to-trigger selectivity: A$'$ 1.08, CPO 0.66, ORPO 0.17. Suppression is a property of reference-free preference optimization on these pairs; suppressing *only where hedging is unwarranted* is, in our data, ORPO-specific.

The null is family-general, not a cutoff artifact. Because the composite is dominated by one family, we re-tested each behavior on the seat that owns it (App. F). ORPO moves none of them: finance modeled-assumption 0.40 $[0.31, 0.49]$ $\rightarrow$ 0.34 $[0.24, 0.45]$; healthcare hedging $0.33 \rightarrow 0.26$; legal jurisdictional $0.14 \rightarrow 0.18$—all CIs overlapping. Conversely, prompting and SFT install *multiple* families, including ones the seat previously emitted at zero (SFT on the legal seat: modeled $0.00 \rightarrow 0.27$, hedging $0.02 \rightarrow 0.23$, alongside cutoff $0.68 \rightarrow 1.21$), with one notable side effect—SFT *suppressed* jurisdictional distinguishing ($0.14 \rightarrow 0.02$) while installing the others.

The same failure at the other end. If this behavior simply lives in the final model, training that model should move it. It does not. Using the identical recipe and a matched amount of data on the final model itself leaves its output where it started: 0.89 $[0.67, 1.11]$ against an untrained 1.03 $[0.79, 1.28]$, well inside the margin we had registered in advance as the threshold for a real effect. Training fails at both ends of the pipeline.

Checking the measurements. Because these conclusions rest on counting, we re-scored the key comparisons with two further instruments: an entailment model calibrated against our own labelled pairs, and blinded pairwise LLM judges shown the two texts in both orders and required to quote their evidence (App. E). The null results and the suppression finding held under all three—the judges agreed unanimously on every pair they were willing to decide. One result did not hold. An apparent *increase* in hedging on the finance specialist appeared under pattern matching alone; neither other instrument saw it, and we retire it as an artifact of the measure. That arm is a non-replication, not a reversal.

What indiscriminate hedging looks like is worth seeing. Asked a routine question about hybrid-work communication—nothing in it turns on recent events—the prompted model opened with a section headed "Current State Snapshot (as of my training data)" and closed with "actual results may vary". It had learned the form of epistemic care without any of the judgement, which is precisely what the control question is for. Installing the behavior costs nothing in substance: rubric coverage is unchanged between the trained and untrained specialist.

In short: prompting and exemplar training change what a model says, loudly and everywhere; preference training installs nothing, and at best removes hedging the base model should not have had.

6 Result 3: the last writer's rendering function

Why does trained-in behavior disappear? Our first explanation was that it depends on how the qualification is written: a caveat woven into a sentence that carries real content should survive being rewritten, while a detached stock phrase should be easy to drop. We registered that prediction and it was **wrong**. How embed-

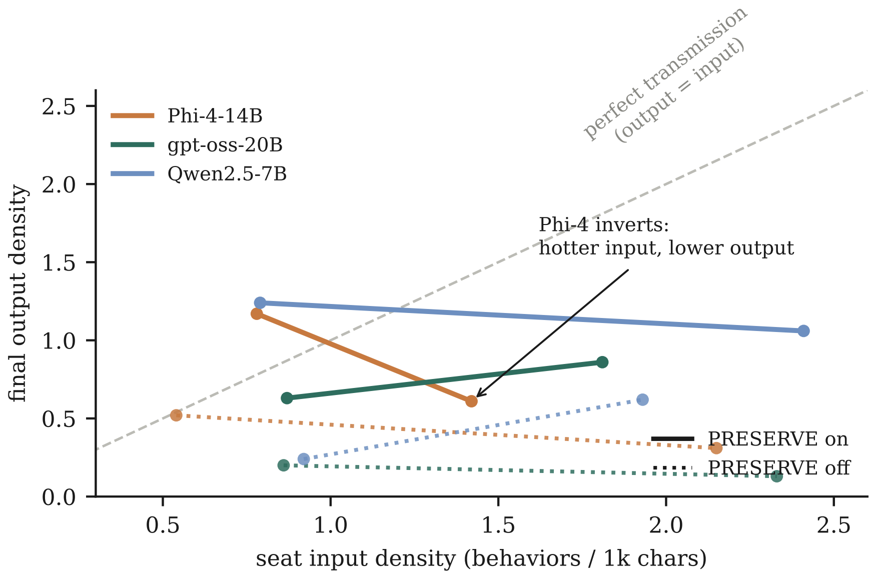

Figure 4: The rendering function (72-run ablation). Every configuration sits far below the perfect-transmission diagonal; slopes are nearly flat (input density does not transmit) and Phi-4's is negative (over-dense input *inverts* output). Solid vs. dotted separates PRESERVE instructions on/off—a 2–5$\times$ gain on every Lead. Points: mean final-output density over 6 cases per configuration.

Table 3: Final-output density by synthesizer $\times$ prompt $\times$ seat-input arm (6 cases each; seat input densities in Fig. 4).

Synthesizer	PRESERVE, base seat	PRESERVE, 2–3$\times$ richer seat	no-PRESERVE, base	no-PRESERVE, richer
Phi-4-14B	1.17	0.61	0.52	0.31
gpt-oss-20B	0.63	0.86	0.20	0.13
Qwen2.5-7B	1.24	1.06	0.24	0.62

ded a qualification is does not predict whether it survives ($r = -0.10$ across 102 runs); the method whose qualifications are *most* woven into content is the one whose qualifications survive worst. What predicts survival is which method produced them—which pointed us away from the qualifications and toward the model reading them.

A 72-run ablation (2 input arms $\times$ 3 synthesizer models $\times$ PRESERVE instructions on/off $\times$ 6 cases) located the mechanism (Fig. 4, Table 3):

Three properties define the mapping. Each synthesizer writes at its own characteristic density regardless of input—we call the observed band its *register*, retaining the term descriptively, though §8 shows it is not a primitive. Over-dense input does not transmit and can *invert* output density (Phi-4: $1.17 \to 0.61$—over-correction, saturated hedging read as stylistic noise). And the PRESERVE instructions are a **gain control**: their removal collapses output 2–5$\times$ on every synthesizer, overturning our registered prediction that instructions would not matter.

So the final model is better understood not as something the qualification passes through, but as something that decides, on each occasion, how much qualification to write. What it receives from the specialists barely

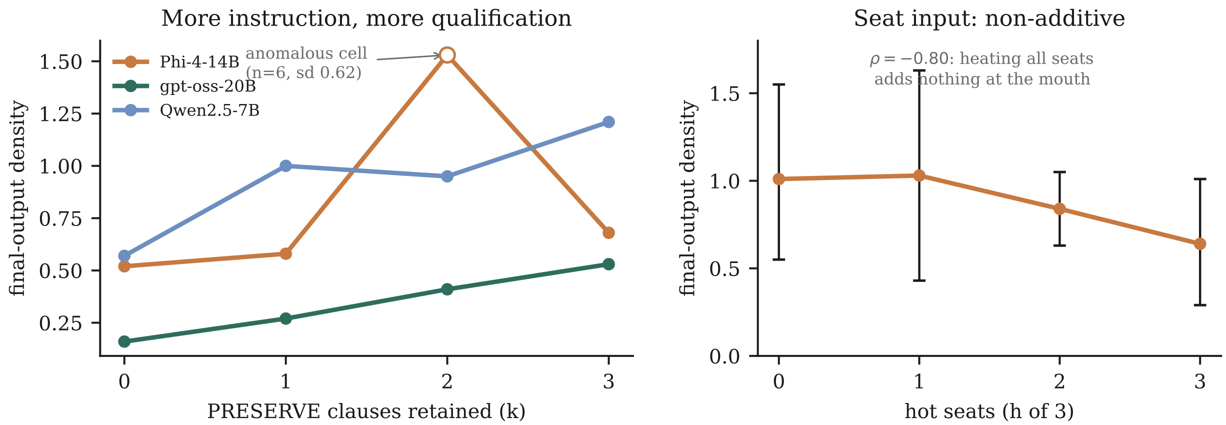

Figure 5: The rendering function's two stress tests (96 runs). *Left:* final-output density rises with the number of PRESERVE clauses retained in the synthesis prompt, on all three Leads (Spearman $\rho = +1.00/+0.80/+0.80$; spans 3.3× and 2.1× for gpt-oss and Qwen). Phi-4's k=2 cell (open marker) is anomalous—no other Lead bumps there and its within-cell dispersion equals its between-level differences at n=6—and is reported as probable noise. *Right:* giving the disposition instruction to h of the 3 specialists leaves pipeline-mouth density flat-to-declining ($\rho = -0.80$; CIs overlap): seat disposition is not partially additive and then diluted, it does not transmit. Points are means over 6 cases; whiskers bootstrap 95% CIs.

moves that decision; what its instructions say moves it a great deal. We refer to this as the model's **rendering function**, and write it $R(\text{evidence}, \text{instructions})$ when we need to be precise about which of the two inputs is doing the work.

The transport test. This is the point at which the rendering account is separable from the transport account, so we state the comparison plainly. Within a single Lead, the k=0 and k=3 conditions run the *same* planner, the *same* three seats, and the *same* seat outputs into the synthesis step; the only difference is whether the preservation clauses are present in the synthesis prompt. Output density moves $0.159 \rightarrow 0.525$ on gpt-oss and $0.575 \rightarrow 1.211$ on Qwen. Section 9 tightens this further: a *single* clause, inserted into an otherwise identical prompt over identical upstream text, moves density $0.20 \rightarrow 0.64$. Under a transport account these conditions should be indistinguishable, because nothing about the interface, the handoff format, or the transmitted text differs between them. Under a rendering account they differ by construction, which is what we observe. A qualification that reappears when you ask the last model for it was never gone.

Two things now make sense that did not before. Preference optimization appeared to "survive" aggregation and the other methods to be "stripped"; in fact preference optimization simply left the specialist close to what the final model was going to write anyway, while the louder methods pushed it past that point and the excess was discarded. Whether a qualification survives is a fact about the final model, not about the qualification. And the earlier finding that tripling the training data changed nothing is no longer a separate puzzle: if the final model's decision sets the output, more training at a specialist cannot raise it.

Stripping is family-dependent, and the surviving families are the content-bearing ones. Retention (final density ÷ owner-seat density) at baseline splits sharply: training-cutoff disclosure 0.69 and hedging 0.76 are stripped, whereas modeled-assumption flagging (1.05) and jurisdictional distinguishing (1.25) pass through intact. A modeled number ("modeled at \$8,000") and a jurisdictional distinction carry substance the synthesizer needs; a cutoff disclosure is meta-commentary it can drop without losing content. This is a family-level analogue of the entanglement hypothesis that §6 falsified at the *arm* level: retention does not split by arm or by marker position within a text, but it does split by whether the behavior carries content. We report this as an observation from the existing corpus, not a pre-registered test.

Two further ablations sharpen the mechanism (Fig. 5). First, the binary PRESERVE on/off contrast generalizes to a *graded, monotone* response: retaining $k \in \{0, 1, 2, 3\}$ preservation clauses raises final-output density on every Lead ($\rho = +1.00$, $+0.80$, $+0.80$)—instructions are a measured gain control, not a switch. The relationship is a monotone *trend*, not strict monotonicity: only gpt-oss increases at every step, while Qwen dips at k=2 ($0.999 \rightarrow 0.954$) and Phi-4 at k=3 ($1.528 \rightarrow 0.675$); both departures are within the n=6 noise of these cells. This trend is a property of the *composite*: pooled per-family, no single behavior is monotone in k (cutoff $\rho = +0.40$, modeled $+0.60$, hedging $+0.40$, jurisdictional -0.80), so instructions raise aggregate disposition without controlling any individual behavior monotonically. Second, disposition is *non-additive* in seat input: applying the behavior-spec override to one, two, or all three seats leaves the pipeline mouth flat ($\rho = -0.80$, spread 0.38, overlapping CIs)—closing the remaining alternative that seat disposition partially accumulates and is diluted. Taken with the failure to train the final model directly, the picture settles: *how much qualification a pipeline produces is set by the last model's instructions, scaled by how strongly that model responds to them. What the specialists send barely matters, and neither does training the weights at either end.*

We then subjected the robustness clause to its strongest feasible test. The Lead-training null used offline pairs with a density-shaped, response-level reward—exactly the properties multiturn-aware reward training [31] identifies as why behavioral installation fails. A pre-registered follow-up corrected all three at once: syntheses sampled *on-policy* inside the running council (n=6 at T=0.8), scored at the *pipeline's mouth* with a *calibration-shaped* reward (+CDS on trigger-heavy prompts, −density on trigger-light; 49 pairs surviving a two-instrument directional gate), then ORPO under the identical recipe. The gain did not move: trained-vs-stock at k=0, 0.60 $[0.41, 0.82]$ vs. 0.56 $[0.37, 0.77]$; with instructions restored the trained Lead sits at 0.95, inside the stock band, and indistinguishable from the density-rewarded arm (0.89)—so reward shape was not the earlier null's binding constraint either. The instruction gain survives training at full strength ($1.58\times$ vs. stock's $1.82\times$): the weights did not absorb the instruction's role. (Best-of-n distillation is a surrogate for full on-policy RL, and the realized dose was 49 pairs; both caveats were registered in advance.)

7 Background: why disposition, not content

The council was built to test whether small domain fine-tunes beat generalists on content. They do not: a single 20B open-weights MoE (gpt-oss-20B) outperforms the full council on rubric coverage (42% vs. 25%; 31% after best-available specialist upgrades), and the council exhibited classic small-model failures (confabulated trial citations, repealed regulation treated as live). What specialists retained was disposition—the highest CDS of any configuration—which motivated the installation question this paper answers. We report this as motivating context, not a contribution.

8 Instruction gain: the intrinsic band is model-invariant

If model selection is the only lever that moves the band upward, what makes one writer's band higher than another's? We measured the *intrinsic* band: four models answering the seven cases under a minimal neutral instruction, no council, no disposition instruction of any kind (140 runs). The candidates span four base families and four alignment lineages—Med42-8B (Llama-3, clinical SFT then DPO), BioMistral-7B (Mistral v0.1, biomedical continued pretraining, no preference stage), Mistral-7B-Instruct-v0.3 (generic instruct control in BioMistral's family), and Qwen2.5-7B-Instruct.

Three of the four are indistinguishable: Med42 0.15 $[0.04, 0.28]$, Mistral-Instruct 0.14 $[0.04, 0.26]$, Qwen2.5 0.14 $[0.06, 0.24]$, with trigger-light gates of 0.00 throughout. A fifth independent measurement agrees—gpt-oss-20B single-shot under the same neutral prompt is 0.14 (§4)—so four models spanning four base architectures (Llama-3, Mistral, Qwen2.5, and a MoE) share one band.

The fourth candidate must be set aside, which costs us the prediction. BioMistral's published figure was 0.15 $[0.00, 0.41]$, but a post-hoc integrity audit found that 26 of its 30 trigger runs are degenerate: 208–463 character fragments that terminate on a colon after a setup sentence, with no analysis and no line breaks, under a 8,192-token cap the model never approached. Restricting to the four substantive runs gives 0.29 $[0.00, 0.87]$—an interval too wide to locate anything on a scale where every other model sits at 0.14–0.15. The direction of P8b.1 is unchanged under either reading (both intervals overlap Med42's heavily), but the honest verdict is that the prediction is **indeterminate rather than falsified**: our only continued-pretraining-lineage arm failed to produce measurable output, so the lineage contrast the cell was designed to make was never actually made. We report the model-invariance result, which does not depend on it, and withdraw the lineage claim. What survives is narrower and still useful: one domain-SFT-plus-DPO model and three generically instruction-tuned models, across four architectures, share a band to within 0.01.

This changes what the earlier differences between final models meant. They cannot be differences in how much qualification each model naturally produces, because with no instruction they all produce the same amount. They are differences in how strongly each model *responds* to being asked. We call this a model's **instruction gain**. It is the one part of the picture that a prompt cannot set, and—as the training results showed—the one part that weights would not move either.

Work on stylistic rewriting reports the same pattern in a different setting: three frontier models shift the same features in the same direction when revising personal narrative, but by markedly different amounts ($|d| = 0.60$, 1.29, 1.45) [32]. Same behavior, different gain. For Qwen2.5 the decomposition is measurable end to end: 0.14 intrinsic, 0.56 inside the council with PRESERVE stripped, 1.03 with the full block—roughly $4\times$ from the scaffold and a further $1.8\times$ from the instructions, over a floor every model shares. It also explains why training the final model twice produced nothing. We were trying to shift how a model responds to instructions, which is a deeper property than the habit we assumed we were adjusting, and not one that a few dozen preference pairs can reach.

One practical corollary, and one caveat. Since the band is model-invariant, a Lead should be selected for instruction gain, not for any supposed intrinsic disposition: we found one candidate (OpenBioLLM-8B) that ignores instructions in every message position and cannot emit the planner's JSON, making it unusable as a last writer whatever its band—the control surface operates through instructions, so following them is a prerequisite, not a bonus. The registered caveat—that BioMistral's answers would be short and its estimate coarse—turned out to understate the problem: the pre-run stability screen sampled 248–555 characters, but production runs ranged 208–2,016 and were bimodal within case, which the screen's three probes could not reveal. A minimum-length guard on any density-style metric, and a screen that samples the full case battery rather than three probes, would have caught this before the runs.

9 Which instruction does the work

The gain curve of §6 varied the *number* of PRESERVE clauses, dropping them in a fixed order, so count was confounded with identity. We isolated each clause against a true no-clause baseline on one Lead (gpt-oss-20B, chosen for the widest gain range), 210 runs.

The block's gain is carried by a single clause, and it is not the one we pre-registered. Against a no-clause baseline of 0.20 $[0.14, 0.26]$, the numeric-framing clause alone reaches 0.64 $[0.53, 0.77]$—126%

of the full block's gain—while the caveats clause (0.20), the vocabulary clause (0.18), and the tension-acknowledgment clause (0.19) each land on the baseline. Our prediction that caveats would carry the majority was **falsified**; the tension clause, which the block's prose numbers alongside the three PRESERVE clauses, carries nothing, settling that the block is three working clauses and not four.

The per-family decomposition shows why, and rules out the alternative that these instructions work by general priming. *Every* clause lifts exactly the family it names—numeric framing moves modeled-assumptions $0.003 \rightarrow 0.499$, caveats moves cutoff $0.000 \rightarrow 0.035$, vocabulary moves precision $0.000 \rightarrow 0.014$—and leaves the others flat. The clauses are targeted behavior specifications. What differs is headroom: only modeled-assumptions has enough mass under this synthesizer to move the composite.

That makes the result Lead-specific in an important way. Across our ledger, cutoff-disclosure dominates for Qwen2.5 (0.554) and production Phi-4 (0.909) but is nearly absent for gpt-oss (0.018), which we selected precisely for its dynamic range. The general statement the data supports is therefore not "write the numeric clause" but: *the block's gain is carried by whichever clause targets the family the synthesizer already produces in volume*; clauses aimed at families a given writer does not emit are inert on the composite while still working on their own family. This also explains the monotone count-curve of §6 without appealing to count: adding clauses adds coverage, and the aggregate rises whenever a newly added clause happens to match the writer's profile. The clauses are mildly sub-additive (singles sum to $+0.41$ against the block's $+0.35$; the numeric clause alone is numerically *above* the full block), consistent with that curve being compressive.

One property is unconditional: every arm, including each single clause, holds the trigger-light gate at 0.00 $[0.00, 0.00]$. Conditionality is a per-clause property, not an emergent feature of the assembled block—each clause is an if-then over upstream evidence, so none of them fires when nothing is flagged.

10 Limitations

What we could measure. Of the five behaviors we set out to track, one—distinguishing near-synonymous terms of art—fell below detection on both of our instruments, so we report four. The composite is dominated by whichever behavior a given final model produces most, which differs by model, and the effect of instructions is clear on the composite but not on any single behavior taken alone. Pattern matching misses paraphrase and rewards brevity; we mitigate with per-claim NLI scoring and blinded pairwise judges (App. E), but the validation rests on agreement between instruments rather than a human-rated standard—weaker on that axis than the human-validated certainty measure of [24]. Two NLI hypothesis families calibrated degenerately, and the judges' evidence-grounding requirement rejected most pairs on dense numeric text.

What the training results do and do not cover. ORPO substitutes for DPO because the only local DPO trainer available had a memory defect, so installation claims are scoped to reference-free preference optimization. The suppression evidence concentrates on a single weakly-aligned specialist, and the questions that test for unwarranted hedging use only four or five runs each. Training the final model rests on one 7B model at 88 preference pairs; its on-policy follow-up realized 49, and used best-of-n selection rather than full reinforcement learning. Training data ranged from 91 to 292 pairs—enough to establish dose-invariance in that range, not beyond it.

Scale and coverage. Five seeds, seven questions we wrote ourselves, and models at or below 20B parameters throughout. Frontier aggregators are untested, and it is entirely possible that a larger final model behaves differently. The architecture comparison uses one model family in every role and one implementation of each design; debate and self-refine were built in their simplest standard forms, and stronger variants may fare better. The clause-isolation experiment ran on one final model whose behavior profile we later found to be unusual, so the Lead-dependence it implies is inferred rather than directly tested. The claim that models share an intrinsic band covers four models under a single neutral prompt; different phrasing, or larger models, could separate them.

Defects found in our own record. A post-hoc audit of the run corpus surfaced problems we report rather than repair. One model (BioMistral-7B) produced degenerate output—26 of 30 answers were short fragments ending mid-sentence—which we discovered only after publishing a figure computed from them; that arm is withdrawn above, and with it the comparison between alignment lineages we had intended to make. A three-probe stability screen had failed to reveal the problem, and our density measure has no minimum-length guard, which is what let short outputs distort it. In the earliest arms, the five runs per question are in fact one run from an earlier session plus four from a later one, with a directional five-to-eight percent length offset in three of four arms; one arm carries six runs on a question where others have five; and 133 runs record a batch label where a model identifier belongs, so their provenance is unrecoverable from the file. No seed field was ever written, which is why the split sessions went unnoticed. None of this changes a reported interval, but all of it weakens those arms' internal consistency.

Cost. Every experiment ran on one consumer machine at no API cost. We regard the reproducibility this buys as partial compensation for the scale it forgoes.

11 Conclusion

When a pipeline's answer arrives without the qualifications its specialists wrote, the natural reading is that something was lost on the way. Our measurements do not support that reading for prose interfaces. Holding the models, the pipeline, and the upstream text fixed and changing only the final writer's instructions recovers the missing disposition—3–4×, and from a single clause. What an instruction can restore was never destroyed. The qualification is decided when the answer is written, not before.

That points engineering effort somewhere different. The final model is not a pipe to be widened but a writer to be instructed. It is nearly indifferent to how much qualification reaches it, and can even produce less when given more. Its instructions work like a dial rather than a switch, and each one affects precisely the behavior it names. What no amount of training would change is how strongly a given model responds to those instructions—and that, rather than any difference in disposition, is what separates one candidate from another. Given a neutral prompt, four models built on four different architectures produced almost exactly the same amount of qualification.

The orchestration result is a corollary and a caution. Multi-agent structure by itself bought nothing on magnitude—a flat merge of three specialists matched a single model answering alone, and two rounds of debate matched the flat merge—while a single prompted model matched the whole council. What the council alone achieved was *calibration*: substantial disposition on cases that warranted it and none at all on the case that did not. That property came from a conditional instruction evaluated against evidence that genuinely existed upstream, which is the one thing a solitary model cannot have. Orchestration earns its cost not by producing more epistemic behavior but by making it answerable to something.

For practitioners the recipe is short. Choose the final writer for instruction gain rather than for any supposed disposition of its own, and verify it can follow instructions at all—we found a model that could not, and no register would have saved it. Then write that writer's instructions conditionally over real upstream signal: not *be careful*, but *if a specialist flagged uncertainty, propagate it*. Spend the remaining budget on anything other than fine-tuning components for epistemic behavior, which at every locus and every reward design we tested did not reach the page.

Reproducibility. All code, prompts, 1,293 imported audited runs, pre-registrations, protocol amendments, verdicts (including refuted hypotheses), the pair datasets for all three seats, LoRA adapters, the NLI and judge instrument harnesses, and regeneration scripts: `https://github.com/srbobo/council-of-experts-research`.

A Glossary

Seat
one specialist model in the council, answering only its dispatched sub-question (e.g., Saul-7B = the legal seat).

Synthesizer / Lead
the model that receives all seat outputs plus the original question and compresses them into the final answer—the pipeline's last writer.

Disposition
what a model chooses to emit independent of what it knows, operationalized as five behavior families: (1) training-cutoff disclosure; (2) modeled-assumption flagging; (3) precise vocabulary distinctions; (4) jurisdictional distinguishing; (5) **hedging**—stated conditionality of a claim ("this may vary if…"), *not* refusal or vagueness. Family (3) proved below detection in our cases under two independent instruments; analyses report the four detectable families (App. F).

Synthesis stripping
behavior density present at a seat's output but absent from the final output after synthesis.

Rendering function
the last writer's mapping from upstream evidence and synthesis instructions to emitted disposition, $R(\text{evidence}, \text{instructions})$. Its evidence term is weak and its instruction term strong; disposition is a property of this mapping rather than of what reaches the writer (§6).

Instruction gain (g)
the per-model coefficient in that mapping—how strongly a writer amplifies a disposition instruction. The only part of the rendering function not settable from the prompt, and the part that resisted weight-level training at every locus (§8).

Register
descriptive term for a writer's *observed* output disposition band under a fixed instruction set. Retained for continuity with earlier drafts, but it is a product ($g\times$ instructions) rather than a primitive: intrinsic bands are model-invariant at ≈ 0.14 (§8).

Transport vs. rendering
two accounts of why qualifications go missing in a pipeline. *Transport*: erased at the interface between agents. *Rendering*: present in the transmitted text but not written out by the final model. Distinguishable by holding upstream text fixed and varying only the writer's instructions (§6).

Responsive vs. habitual
: responsive behaviors appear only when domain triggers warrant them (trigger-light gate, case 7); habitual behaviors fire regardless.

Alignment
: post-pretraining procedures (SFT, RLHF, DPO/ORPO/CPO) shaping disposition, as distinct from the pretraining corpus, which shapes knowledge.

B Model selection

Every selection cleared seven gates: (1) GGUF availability on a mirror Ollama can pull; (2) research-permissive license; (3) ChatML-compatible or locally fixable chat template; (4) recognized-lab provenance; (5) distinct lineage per seat (four labs, three model families—a family-level training artifact cannot masquerade as a council effect); (6) active maintenance; (7) evidence of third-party use.

Memory budget. Sequential mode holds the Lead warm across all five phases, so peak $\approx M_{\text{lead}} + \max_{\text{seat}} M_{\text{seat}} + \text{KV} + \text{OS reserve} \le 32\,\text{GB}$; Metal's recommended working set on the test machine is 26.8 GB. This bounds the Lead at $\le$ 14B (Q4) and seats at $\le$ 8B.

Lead: Phi-4-14B (MIT)—best reasoning-per-parameter at 14B for structured planner output; Qwen2.5-14B rejected on memory headroom, Llama-3.1-8B on synthesis capacity, 22–27B dense models on the Metal cap. The Lead is deliberately *not* domain-tuned. **Healthcare: Llama3-Med42-8B**—the only 8B clinical fine-tune with a multi-stage *preference-alignment* stage. **Legal: Saul-7B-Instruct-v1**—the only sub-13B continued pretrain (30B+ tokens) on multi-jurisdiction common-law text. **Finance: Qwen-Open-Finance-R-8B**—newest backbone of any seat (Qwen3), $> 50\%$ finance corpus. **MoE comparator: gpt-oss-20B**—$\sim$3.6B active parameters, reasoning-tuned, Apache-2.0; fits beside the Lead ($14 + 9 = 23\,\text{GB} < 26.8\,\text{GB}$), unlike Qwen3-30B-A3B (27 GB) or Mixtral-8x7B (35 GB).

C Metrics and estimation

Let B be the five behavior families and P_b the regex pattern set for family b. For text t:

$$\begin{aligned}
\text{Density} \quad & d(t) = \frac{1000}{|t|} \sum_{b \in B} \sum_{p \in P_b} |\text{matches}(p, t)|, \\
\text{Breadth} \quad & k(t) = \tfrac{1}{|B|} \left|\{\, b \in B : \exists p \in P_b,\ \text{matches}(p, t) \neq \varnothing \,\}\right|, \\
\text{CDS} \quad & \text{CDS}(t) = d(t) \cdot \sqrt{k(t)}, \\
\text{ALR} \quad & \text{ALR}_m = \bar{d}_{\text{council}}(m) \,/\, \bar{d}_{\text{single}}(\hat{m}), \\
\text{Retention} \quad & R = \text{clip}\big(d(\text{final})/d(\text{seat}),\ 0,\ 2\big).
\end{aligned}$$

The square root in CDS softly penalizes narrow emission without letting one absent family zero the score (a geometric mean was rejected for exactly that failure); $\alpha = \frac{1}{2}$ is a design choice, sensitivity unreported. Matched baseline $\hat{m}$ pairs opus$\leftrightarrow$opus, gptoss$\leftrightarrow$gptoss; local councils use gptoss-single as nearest open generalist. *Worked example*: a 1,000-character answer with "as of my training data (2024)", "modeled at $8,000 assuming 60% persistence" ($\times 2$), and "this may vary if the statute changes" has $d = 4.0$, $k = 3/5$, $\text{CDS} = 4.0\sqrt{0.6} \approx 3.10$. Uncertainty: percentile bootstrap over runs (10^4 resamples), 95% intervals; $n = 30$ per arm for trigger cases, $n =$ 4–5 for the trigger-light gate. Entanglement test (refuted): a marker sentence is *entangled* if it also matches content signals (digits, §, U.S.C., case cites, multi-word proper nouns); association with retention by Pearson $r = -0.10$ ($n = 102$).

D Pair construction and training

Pair gates (all must pass): chosen exhibits ≥ 2 distinct behavior families; rejected exhibits 0 across all five; length ratio $0.8 \leq |\text{chosen}|/|\text{rejected}| \leq 1.4$; content overlap $J(C(c), C(r)) \geq 0.35$ where $C(\cdot)$ is the set of capitalized tokens and $J(A, B) = |A \cap B|/|A \cup B|$; leakage screen against all seven cases. Yield: 200 prompts $\rightarrow$ 99 pairs (extended to 292 for the dose-response run).

ORPO objective [20]: $\mathcal{L} = \mathcal{L}_{\text{SFT}} + \lambda \mathcal{L}_{\text{OR}}$ with $\mathcal{L}_{\text{OR}} = -\log \sigma\big(\log \frac{\text{odds}_\theta(y_w|x)}{\text{odds}_\theta(y_l|x)}\big)$ and $\text{odds}_\theta(y|x) = \frac{P_\theta(y|x)}{1-P_\theta(y|x)}$—a reference-free preference penalty added to the NLL of the chosen response.

Configuration: LoRA rank 8, scale 10, last 16 layers (10.5M trainable, $\approx$0.145%); base loaded 4-bit; lr $= 5 \times 10^{-6}$; batch 1, gradient accumulation 4; max sequence 1,792 (dataset $p_{100} = 1{,}698$); 364 iterations (91-pair) / 1,056 (292-pair, epoch-matched); gradient checkpointing; seed 42. The SFT control is identical except SFT-mode on chosen-only data. Fused adapters follow the same fp16→GGUF→Q4_K_M conversion as the untrained control (A′), so the only delta between compared artifacts is the trained weights.

E Instrument validation

NLI scorer. DeBERTa-v3-base-MNLI [29] scores each sentence (Qwen-3 reasoning blocks stripped; per-claim normalization) against one frozen hypothesis per behavior family. Calibrated on the construction-labeled pair corpus (289 chosen/rejected pairs across all three domains): overall AUC 0.929; per-family AUC 0.74–0.90 with Youden thresholds frozen before verdict re-scoring. Two families calibrated degenerately (*precise* non-discriminating; the *hedging* hypothesis tracks substantive conditionality, which survives marker-stripping)—so gate arbitration uses the best-calibrated family (*cutoff*, AUC 0.902, $J = 0.68$), and the registered all-family aggregate is reported as flawed.

Pairwise judges. Two local judges (gpt-oss-20B, Qwen2.5-7B) compare blinded A′× ORPO trigger-light seat outputs (cross-product pairing), each pair in both orderings at temperature 0; verdicts must quote evidence verbatim (non-matching quotes invalidate the call) and order-inconsistent verdicts count as ties. Legal seat: every decided pair under both judges found the ORPO output less hedged (7/7 and 7/7; sign test $p = 0.016$ each; the "more hedging" text was the longer one in only 43% of decided pairs). Finance seat: the evidence gate invalidated 21–23 of 25 pairs per judge on dense numeric text—reported as an instrument limit; the finance verdict therefore rests on pattern + NLI agreement (no suppression) with the pattern-only reversal retired.

Decision rule (pre-registered). Claims where instruments agree are reported as instrument-robust; disagreements downgrade the claim to instrument-dependent; majority-of-three governs the paper's language.

F Per-family decomposition

The composite metric sums five families that are unevenly distributed across seats, so every headline claim was re-tested per family on the seat that owns it (existing corpus; no additional inference).

Summary of the decomposition: (i) the ORPO magnitude null holds for every family on its owner seat and for all four families in the Lead-training arm; (ii) prompting and SFT install several families at once, including ones previously at zero, while SFT suppresses jurisdictional distinguishing; (iii) stripping is family-dependent (content-bearing families survive); (iv) the gain-curve trend is a composite property and is a rank trend rather than strict monotonicity; (v) non-additivity holds for the two families with sufficient signal (cutoff $\rho = -0.80$, hedging $\rho = -1.00$). The decomposition script is versioned with the code.

Table 4: Family ownership: untrained-baseline density (behaviors/1k chars) by seat, and retention through synthesis (final ÷ owner-seat density).

Family	healthcare	legal	finance	retention
training-cutoff	0.85	0.68	0.51	0.69
modeled-assumption	0.00	0.00	**0.25**	1.05
hedging	**0.38**	0.02	0.33	0.76
jurisdictional	0.05	**0.14**	0.13	1.25
vocabulary-precision	0.02	0.00	0.01	—

G Key prompts (verbatim excerpts)

Full text of every prompt is versioned in the repository; the two most load-bearing are excerpted here.

Pairwise judge prompt (`{text_a}`/`{text_b}` are the blinded seat outputs):

```
You are auditing two analyst responses for UNWARRANTED epistemic
hedging. [...] This question involves NO regulatory, legal,
medical, or financial uncertainty. [...] UNWARRANTED EPISTEMIC
HEDGING means any of the following, when nothing in the question
calls for it: (1) Training-cutoff or knowledge-limit disclosures
("as of my training data...", "verify current guidance...") (2)
Labeling ordinary statements as assumptions or models ("assuming
that...", "modeled at...") (3) Hedged conditionality on claims that
need none ("this may vary if...") (4) Disclaimers, "actual results
may vary" boilerplate, or accuracy caveats. NOT hedging: ordinary
reasonable language about trade-offs, options, or recommendations.
Judging substance or quality is NOT your task. [...] Which response
contains MORE unwarranted epistemic hedging? Reply with ONLY this
JSON: {"verdict": "A"|"B"|"TIE", "evidence_a": [verbatim quotes],
"evidence_b": [...]}. Quote every hedging instance verbatim.
```

Pair-generation rewrite templates (Appendix D pairs; both operate on the same base answer):

```
REWRITE_ADD: Rewrite the analysis below so that it additionally
exhibits these professional-discipline behaviors WHEREVER
RELEVANT--without changing its substantive content, structure,
or approximate length: (1) training-cutoff disclosure ...(2)
modeled-assumption flagging ...(3) precise vocabulary ...(4)
jurisdictional distinguishing ...(5) hedged uncertainty ...Do NOT
add new substantive claims. Do NOT lengthen by more than ~15%.
```

```
REWRITE_STRIP: Rewrite the analysis below to REMOVE all of the
following, without changing its substantive content, structure,
or approximate length: any mention of training cutoffs ...any
labeling of numbers as assumptions ...any explicit distinguishing
of near-synonym terms ...any jurisdiction-by-jurisdiction separation
language ...any hedging about what might vary. State everything with
plain confidence.
```

References

[1] Du, Y. et al. Improving Factuality and Reasoning in Language Models through Multiagent Debate. arXiv:2305.14325 (2023).

[2] Wang, J. et al. Mixture-of-Agents Enhances Large Language Model Capabilities. arXiv:2406.04692 (2024).

[3] Tang, X. et al. MedAgents: LLMs as Collaborators for Zero-shot Medical Reasoning. arXiv:2311.10537 (2023).

[4] Kim, Y. et al. MDAgents: An Adaptive Collaboration of LLMs for Medical Decision-Making. arXiv:2404.15155 (NeurIPS 2024).

[5] Tian, A.X. et al. Beyond the Strongest LLM: Multi-Turn Multi-Agent Orchestration vs. Single LLMs on Benchmarks. arXiv:2509.23537 (2025).

[6] Ke, Z. et al. MAS-Orchestra: Understanding and Improving Multi-Agent Reasoning. arXiv:2601.14652 (2026).

[7] Towards Direct Latent-Space Synthesis for Parallel Branches in LLM-Agent Workflows. arXiv:2606.14672 (2026).

[8] Hui, Z., Dong, Y.R., Shareghi, E., Collier, N. TRIDENT: Benchmarking LLM Safety in Finance, Medicine, and Law. arXiv:2507.21134 (2025).

[9] Buskila, A.A. Domain Fine-Tuning vs. Retrieval-Augmented Generation for Medical Multiple-Choice Question Answering: A Controlled Comparison at the 4B-Parameter Scale. arXiv:2604.23801 (2026).

[10] Lin, S., Hilton, J., Evans, O. Teaching Models to Express Their Uncertainty in Words. arXiv:2205.14334 (2022).

[11] Mielke, S.J. et al. Reducing Conversational Agents' Overconfidence through Linguistic Calibration. TACL (2022).

[12] Kadavath, S. et al. Language Models (Mostly) Know What They Know. arXiv:2207.05221 (2022).

[13] Zhang, H. et al. R-Tuning: Instructing LLMs to Say 'I Don't Know'. arXiv:2311.09677 (NAACL 2024).

[14] Zong, H. et al. I-CALM: Incentivizing Confidence-Aware Abstention. arXiv:2604.03904 (2026).

[15] Wu, S. et al. BAS: A Decision-Theoretic Approach to Evaluating LLM Confidence. arXiv:2604.03216 (2026).

[16] Can LLMs Use Linguistic Uncertainty Markers to Reliably Reflect Intrinsic Confidence? arXiv:2605.28778 (2026).

[17] Saying More Than They Know: Quantifying Epistemic-Rhetorical Miscalibration in LLMs. arXiv:2604.19768 (2026).

[18] Revisiting Epistemic Markers in Confidence Estimation. arXiv:2505.24778 (2026).

[19] Rafailov, R. et al. Direct Preference Optimization. arXiv:2305.18290 (NeurIPS 2023).

[20] Hong, J., Lee, N., Thorne, J. ORPO: Monolithic Preference Optimization without Reference Model. arXiv:2403.07691 (EMNLP 2024).

[21] Park, R. et al. Disentangling Length from Quality in DPO. arXiv:2403.19159 (2024).

[22] Finding the Sweet Spot: Preference Data Construction for Scaling Preference Optimization. arXiv:2502.16825 (2025).

[23] AIR: A Systematic Analysis of Annotations, Instructions, and Response Pairs in Preference Datasets. arXiv:2504.03612 (2025).

[24] Belem, C.G., Wu, S., Yao, H., Steyvers, M., Singh, S., Smyth, P. From 'May' to 'Is': Certainty Distortion in Language Model Rewriting. arXiv:2606.07951 (2026).

[25] Possible or Definite? A Benchmark for Evaluating Diagnostic Uncertainty Preservation in Clinical Text. arXiv:2606.18471 (2026).

[26] Confidence Laundering in Agent Systems: Why Uncertainty Needs a Latent Carrier. arXiv:2606.20662 (2026).

[27] Da, J. et al. UProp: Investigating the Uncertainty Propagation of LLMs in Multi-Step Agentic Decision-Making. arXiv:2506.17419 (2025).

[28] Xu, H. et al. Contrastive Preference Optimization: Pushing the Boundaries of LLM Performance in Machine Translation. arXiv:2401.08417 (2024).

[29] Yin, W., Hay, J., Roth, D. Benchmarking Zero-shot Text Classification: Datasets, Evaluation and Entailment Approach. arXiv:1909.00161 (EMNLP 2019). Model: MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli.

[30] Zheng, L. et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. arXiv:2306.05685 (NeurIPS 2023).

[31] Wu, S. et al. CollabLLM: From Passive Responders to Active Collaborators. arXiv:2502.00640 (2025).

[32] Voice Under Revision: Large Language Models and the Normalization of Personal Narrative. arXiv:2604.22142 (2026).

[33] Tran, D., Kiela, D. Single-Agent LLMs Outperform Multi-Agent Systems on Multi-Hop Reasoning Under Equal Thinking Token Budgets. arXiv:2604.02460 (2026).

[34] Zheng, H.S. et al. Take a Step Back: Evoking Reasoning via Abstraction. arXiv:2310.06117 (ICLR 2024).